



Assignment 02

Submitted By	Ijlal Ahmad
Submitted To	Ms. Saira Kishwar
Roll No	FL-21565
Date	05-05-2025
Subject	Web Engineering

Data Types in JavaScript

JavaScript has 8 Datatypes

1. **String:** Textual data, can use single quotes, double quotes, or backticks
2. **Number:** Both integers and decimals, includes special values like `Infinity` and `NaN`
3. **BigInt:** For integers larger than $2^{53} - 1$ (`Number.MAX_SAFE_INTEGER`)
4. **Boolean:** Only `true` or `false` values
5. **Undefined:** Variables declared but not initialized
6. **Null:** Intentional absence of value (must be assigned)
7. **Symbol:** Unique identifiers (even with same description they're different)
8. **Object:** Collections of properties (includes arrays, functions, dates)

Examples

String - represents textual data
`let username = "JohnDoe123";`

Number - represents both integer and floating-point numbers
`let age = 30;`
`let temperature = 36.6;`

BigInt - represents integers of arbitrary length
`let bigNumber = 9007199254740991n; // Appending 'n' creates a BigInt`
`let hugeNumber = BigInt("123456789012345678901234567890");`

Boolean - represents logical values true/false
`let isLoggedIn = true;`

```
let hasPermission = false;
```

Undefined - represents a variable that has been declared but not assigned

```
let unassignedVariable;  
let undefinedVariable = undefined;
```

Null - represents intentional absence of any object value

```
let emptyValue = null;
```

Symbol - represents unique, immutable primitive values

```
let id = Symbol("unique identifier");  
let anotherId = Symbol("unique identifier"); // Different from id above  
const COLOR_RED = Symbol("red");
```

Object - represents collections of key-value pairs

```
let person = {  
  name: "Alice",  
  age: 28,  
  isAdmin: false  
};
```

Operators in JavaScript

Javascript operators are used to perform different types of mathematical and logical computations.

There are different types of JavaScript operators:

- Arithmetic Operators
- Assignment Operators
- Comparison Operators
- String Operators
- Logical Operators
- Bitwise Operators
- Ternary Operators
- Type Operators

JavaScript Arithmetic Operators

Arithmetic Operators are used to perform arithmetic on numbers

Arithmetic Operators Example

```
let a = 3;  
let x = (100 + 50) * a;
```

Operator	Description
+	Addition
-	Subtraction
*	Multiplication

**	Exponentiation (ES2016)
/	Division
%	Modulus (Division Remainder)
++	Increment
--	Decrement

JavaScript Assignment Operators

Assignment operators assign values to JavaScript variables.

Assignment Example

```
let x = 10;
x += 5;
```

Operator	Example	Same As
=	x = y	x = y

<code>+=</code>	<code>x += y</code>	<code>x = x + y</code>
<code>-=</code>	<code>x -= y</code>	<code>x = x - y</code>
<code>*=</code>	<code>x *= y</code>	<code>x = x * y</code>
<code>/=</code>	<code>x /= y</code>	<code>x = x / y</code>
<code>%=</code>	<code>x %= y</code>	<code>x = x % y</code>
<code>**=</code>	<code>x **= y</code>	<code>x = x ** y</code>

JavaScript Comparison Operators

Operator	Description
<code>==</code>	equal to

===	equal value and equal type
!=	not equal
!==	not equal value or not equal type
>	greater than
<	less than
>=	greater than or equal to
<=	less than or equal to
?	ternary operator

Comparison Operator Example

```
10 = 10;  
5 != 10;  
15 > 10;  
5 < 10;
```

JavaScript String Comparison

All the comparison operators above can also be used on strings

Example

```
let text1 = "A";  
let text2 = "B";  
let result = text1 < text2;
```

JavaScript String Addition

The **+** can also be used to add (concatenate) strings

Example

```
let text1 = "John";  
let text2 = "Doe";  
let text3 = text1 + " " + text2;  
Result=John Doe
```

JavaScript Logical Operators

Operator	Description
&&	logical and
	logical or
!	logical not

Example

Logical and (&&)

```
if (age >= 18 && hasID) {  
  console.log("Access granted");  
} else {  
  console.log("Access denied");  
}
```

Logical or (||)

```
if (isWeekend || isHoliday) {  
  console.log("No work today");  
} else {  
  console.log("Go to work");  
}
```

Logical not (!)

```
if (!isLoggedIn) {  
  console.log("Please log in");  
} else {  
  console.log("Welcome back!");  
}
```

JavaScript Bitwise Operators

Bit operators work on 32 bits numbers.

Any numeric operand in the operation is converted into a 32 bit number. The result is converted back to a JavaScript number.

Operator	Description	Example	Same as	Result	Decimal
&	AND	5 & 1	0101 & 0001	0001	1
	OR	5 1	0101 0001	0101	5
~	NOT	~ 5	~0101	1010	10
^	XOR	5 ^ 1	0101 ^ 0001	0100	4
<<	left shift	5 << 1	0101 << 1	1010	10
>>	right shift	5 >> 1	0101 >> 1	0010	2
>>>	unsigned right shift	5 >>> 1	0101 >>> 1	0010	2

